GRADE 7 ACCELERATED · UNIT 4

Unit 4 Review

Solving Problems with Rational Numbers **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Adding and Subtracting in Scientific Notation

Lesson 1 · MA.8.NSO.1.5 · 6 questions

1. $(4.1\times10^{6})+(2.3\times10^{6})$

6.4×10^{6}

Same power: add the coefficients, copy the power down.

2. $(7.6\times10^{-4})-(2.9\times10^{-4})$

4.7×10^{-4}

3. $(8.2\times10^{5})+(4.9\times10^{5})$

1.31×10^{6}

13.1×10^{5} *is the value and not the finished form.*

4. $(5.4\times10^{8})+(7.1\times10^{7})$

6.11×10^{8}

$7.1\times10^{7}=0.71\times10^{8}$.

5. $(6.3\times10^{-2})-(8\times10^{-4})$

6.22×10^{-2}

$8\times10^{-4}=0.08\times10^{-2}$ *; the exponents are two apart, the widest gap allowed.*

6. Lena writes $(3.2\times10^{4})+(5.5\times10^{6})=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

She added the coefficients without first writing both over the same power of ten, and she added the exponents. The sum is 5.532×10^{6} **.**

$3.2\times10^{4}=0.032\times10^{6}$ *, and* $5.5+0.032=5.532$.

PART B. Multiplying and Dividing in Scientific Notation

Lessons 2–3 · MA.8.NSO.1.5 · 6 questions

7. $(4\times10^{3})(2\times10^{6})$

8×10^{9}

Multiply the coefficients, add the exponents.

8. $\frac{7.2\times10^{8}}{1.8\times10^{3}}$

4×10^{5}

Divide the coefficients, subtract the exponents, top minus bottom.

9. $(5\times10^{6})(8\times10^{-2})$

4×10^{5}

40×10^{4} *is not finished.*

10. $\frac{2.4\times10^{5}}{8\times10^{9}}$

3×10^{-5}

0.3×10^{-4} *is not finished either.*

11. $\frac{4\times10^{7}}{4\times10^{-3}}$

1×10^{10}

$7-(-3)=10$ *. Subtracting a negative goes up.*

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^{2} seconds?

2.6×10^{17} ***photons***

$6.5\cdot4=26$ *, so* 26×10^{16} *before it is finished.*

PART C. Real-World Problems in Scientific Notation

Lesson 4 · MA.8.NSO.1.6 · 4 questions

13. A rectangular panel measures 3.2×10^{2} mm by 1.45×10^{2} mm. Find its area, in scientific notation. Do not round.

4.64×10^{4} ***square millimeters***

$3.2\cdot1.45=4.64$ *, and* $10^{2}\cdot10^{2}=10^{4}$ *.*

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

8×10^{1} ***km/h***

$\frac{4.8\times10^{2}}{6}=0.8\times10^{2}$ *, which is not finished.*

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

5.68×10^{3} ***mm***

$2.8\times10^{2}=0.28\times10^{3}$ *, and* $5.4+0.28=5.68$.

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

7.258×10^{-2} ***g***

$3.42\times10^{-3}=0.342\times10^{-2}$ *, and* $7.6-0.342=7.258$.

PART D. Evaluating Expressions with Radicals

Lesson 5 · MA.8.NSO.1.7 · 7 questions

17. $\sqrt{169}$

13

18. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

19. $\sqrt{7^{2}-24}$

5

Finish under the bar: 49 − 24 = 25.

20. $\sqrt[3]{\frac{1}{27}}$

$\frac{1}{3}$

Top and bottom each get the root.

21. $\sqrt{100}\cdot4^{-2}$

$\frac{5}{8}$

$10\cdot\frac{1}{16}$.

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2+7}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

No. A cube root asks for the number used three times, and $(-4)(-4)(-4) = -64$ ***, so the value is −4. It is a square root of a negative number that is not a real number.***

This is the first misconception the state guide names for this benchmark.

PART E. The Order of Operations

Lesson 6 · MA.8.NSO.1.7 · 7 questions

24. $24 - 3 \cdot 5 + 1$

10

Multiply first, then left to right.

25. $48 \div 6 \cdot 2$

16

One level, left to right: 8 · 2. Not 4.

26. $4 + 3(8-5)^2$

31

$4 + 3(9)$.

27. $\frac{6^2+9}{3^2-4}$

9

45 over 5.

28. $-6^2 + (-6)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

9

$20 \cdot \frac{9}{20}$.

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ● **B.** 102
- ○ C. 42
- ○ D. 150

B

PART F. Real-World Problems and the Order of Operations *Lessons 7–8 · MA.8.NSO.1.7 · 5 questions*

31. Evaluate $5x^2$ when $x=-3$.

45

$5(-3)^2=5(9)$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

9,800 joules

$200(49)$ *. The unit is part of the answer.*

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

78 dollars and 364 dollars

$30+12(4)=78$ *and* $30(12)+4=364$ *. The same three numbers and the same words; what decides is which amount is repeated.*

34. SSDD. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery. Same surface, four different questions: a measurement, an expression, an evaluation, and a size read back out of a total. Only part b has no number for an answer, and only part d runs the structure backwards.

a. What is the area of the patio?

225 square feet

b. Write an expression for the total cost.

$15^2(2.20)+60$

The rate multiplies the area; the flat fee does not.

c. Find the total cost.

555 dollars

$225(2.20)=495$ *, then* $+60$.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

20 feet

$940-60=880$ *, and* $880\div2.20=400$ *, whose square root is 20.*

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50 + 140)$
- ○ B. $18(3.50) + 140$
- ○ C. $18^2 + 3.50 + 140$
- ● **D.** $18^2(3.50) + 140$

D